A PRELIMENERY REPORT ON

Evaluate performance enhancement of parallel Quicksort Algorithm using MPI.

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SUBMITTED BY

STUDENT NAME	Exam No
Pushkar Sanjay Poman	B22069

DEPARTMENT OF COMPUTER ENGINEERING

Zeal College of Engineering & Research

NARHE, PUNE - 411041

SAVITRIBAI PHULE PUNE UNIVERSITY

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CERTIFICATE

This is to certify that the project report entitles

"Evaluate performance enhancement of parallel Quicksort Algorithm using MPI"

Submitted by

STUDENT NAME	Exam No
Pushkar Snajay Poman	B22069

is a bonafide student of this institute and the work has been carried out by him/her under the supervision of **Prof.Nitisha K. Rajgure** and it is approved for the fulfillment of the requirement of Savitribai Phule Pune University, for the award of the degree of Bachelor of Engineering (Computer Engineering).

Prof.Nitisha K. Rajgure
Guide
Department of Computer Engineering

Prof. A. V. Mote
Head
Department of Computer Engineering

Dr. Ajit M. Kate
Principal,
Zeal College of Engineering & Research, Pune-41

Place: Pune
Date:

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PUSHKAR SANJAY POMAN

ABSTRACT

In today's computing era, efficiently sorting large datasets is essential for improving system performance and reducing execution time. Among various sorting techniques, Quicksort is widely used due to its average-case efficiency and simplicity. However, traditional sequential Quicksort does not fully utilize the capabilities of modern multi-core and distributed systems. This project focuses on implementing both standard sequential Quicksort and a parallel version of Quicksort using the Message Passing Interface (MPI), and comparing their performance in terms of execution time and efficiency.

The primary problem addressed in this project is the limited performance of sequential sorting algorithms when dealing with large datasets and the underutilization of available processing resources. To overcome this limitation, the project explores the parallelization of the Quicksort algorithm using MPI, where the dataset is divided among multiple processes that execute sorting tasks concurrently. This approach distributes the workload across multiple processors, thereby reducing the overall computation time.

Experimental analysis is conducted to compare both approaches based on execution time, scalability, and efficiency under different input conditions. The results demonstrate that the MPI-based parallel Quicksort significantly improves performance for large datasets by leveraging parallel processing. The study concludes that parallel Quicksort using MPI is more efficient than its sequential counterpart in distributed computing environments.

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INTRODUCTION

1.1 Overview

Sorting is a fundamental operation in computer science, widely used in searching, data analysis, and optimization tasks. Quicksort is one of the most efficient sorting algorithms, based on the divideand-conquer approach, with an average time complexity of O(n log n). However, traditional Quicksort operates sequentially and does not take advantage of modern multi-core or distributed systems.

To enhance performance, parallel computing techniques such as MPI can be used to divide the sorting task among multiple processes. This project focuses on implementing both sequential Quicksort and MPI-based parallel Quicksort, and analyzing their performance in terms of execution time and efficiency.

1.2 Problem Definition and Objectives

Problem Definition:

Sequential Quicksort algorithms do not utilize multiple processors, leading to increased execution time when handling large datasets.

Objectives:

- To implement standard (sequential) Quicksort
- To develop a parallel Quicksort using MPI
- To compare execution time and efficiency of both approaches
- To analyze performance for different input sizes and scenarios.

REPORT

Implementation Overview

The project is implemented in C++ using MPI to compare the performance of sequential Quicksort and parallel Quicksort. MPI enables communication between multiple processes, allowing distributed execution of sorting tasks.

The main objective is to evaluate how parallel processing improves sorting performance. The program measures execution time and demonstrates how workload distribution among processes enhances efficiency.

Implementation Details

Programming Language Used:

C++ with MPI library

Libraries Used:

- mpi.h – for MPI communication
- iostream – for input/output operations
- vector – for dynamic array handling
- algorithm – for sorting and merging

Quicksort Function:

- Implements divide-and-conquer strategy
- Selects a pivot element
- Partitions array into smaller elements and larger elements
- Recursively sorts partitions

Sequential Quicksort:

- Executes sorting on a single process
- Time complexity: O(n log n) (average case)
- Does not utilize multiple processors

Parallel Quicksort using MPI:

- Data is divided among multiple processes
- Each process sorts its portion independently
- MPI_Scatter distributes data
- MPI_Gather collects sorted results
- Final merging is performed

Main Execution:

- Initializes MPI environment
- Generates random dataset
- Distributes data among processes
- Performs local sorting
- Collects and merges results
- Measures execution time using MPI_Wtime()

Performance Evaluation:

- Sequential Quicksort executes on a single process and shows consistent performance
- Parallel Quicksort distributes workload across multiple processes
- Significant improvement observed for large datasets
- Communication overhead exists but is outweighed by parallel speedup

System Requirements:

- C++/WSL installed
- Multi-core processor for multiprocessing benefits
- Basic terminal or IDE (VS Code, PyCharm, or IDLE)

Implementation Code

```
#include <mpi.h>
#include <iostream>
#include <vector>
#include <algorithm>

using namespace std;

// Sequential quicksort
void quicksort(vector<int>& arr, int low, int high) {
    if  (low  <  high)  {  int
        pivot = arr[high];
        int i = low - 1;

        for (int j = low; j < high; j++) {
            if (arr[j] < pivot) {
                i++; swap(arr[i],
                arr[j]);
            }
        }

        swap(arr[i + 1], arr[high]); int
        pi = i + 1;

        quicksort(arr, low, pi - 1); quicksort(arr,
        pi + 1, high);
    }
}

int main(int argc, char* argv[]) {
    int rank, size;

    MPI_Init(&argc, &argv);
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    int n = 16; // total elements vector<int>
    data;

    if (rank == 0) {
```

```
        cout << "Original array: "; for (int
        i = 0; i < n; i++) {
        data.push_back(rand() % 100);
        cout << data[i] << " ";
        } cout <<
        endl;
    }

    // Scatter data to all processes int
    local_n = n / size;
    vector<int> local_data(local_n);

    MPI_Scatter(data.data(), local_n, MPI_INT, local_data.data(),
                local_n, MPI_INT,
                0, MPI_COMM_WORLD);

    // Local sort
    quicksort(local_data, 0, local_n - 1);
    // Gather sorted subarrays
    vector<int> gathered; if
    (rank == 0) {
        gathered.resize(n);
    }

    MPI_Gather(local_data.data(), local_n, MPI_INT, gathered.data(),
               local_n, MPI_INT,
               0, MPI_COMM_WORLD);

    // Final merge at root if
    (rank == 0) {
        sort(gathered.begin(), gathered.end());

        cout << "Sorted array: "; for
        (int x : gathered)
            cout << x << " ";
        cout << endl;
    }

    MPI_Finalize();
    return 0;
}
```

RESULTS

3.1 Outcomes

- Successfully implemented sequential and parallel Quicksort
- Sequential Quicksort performed efficiently for small datasets
- Parallel Quicksort showed improved performance for large datasets
- MPI effectively utilized multiple processes
- Performance improvement increases with dataset size
- Parallel approach reduces overall execution time

3.2 Screenshots

Welcome × mpi_quicksort.cpp ×

mpi_quicksort.cpp

```cpp
#include <mpi.h>
#include <iostream>
#include <vector>
#include <algorithm>

using namespace std;

// Sequential quicksort
void quicksort(vector<int>& arr, int low, int high) {
    if (low < high) {
        int pivot = arr[high];
        int i = low - 1;

        for (int j = low; j < high; j++) {
            if (arr[j] < pivot) {
                i++;
                swap(arr[i], arr[j]);
            }
        }

        swap(arr[i + 1], arr[high]);
        int pi = i + 1;

        quicksort(arr, low, pi - 1);
        quicksort(arr, pi + 1, high);
    }
}

int main(int argc, char* argv[]) {
```

Ln 65, Col 52 Spaces: 4 U

Welcome mpi_quicksort.cpp ×

mpi_quicksort.cpp

```cpp
int main(int argc, char* argv[]) {
    int rank, size;

    MPI_Init(&argc, &argv);
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    int n = 16; // total elements
    vector<int> data;

    if (rank == 0) {
        cout << "Original array: ";
        for (int i = 0; i < n; i++) {
            data.push_back(rand() % 100);
            cout << data[i] << " ";
        }
        cout << endl;
    }

    // Scatter data to all processes
    int local_n = n / size;
    vector<int> local_data(local_n);

    MPI_Scatter(data.data(), local_n, MPI_INT,
                local_data.data(), local_n, MPI_INT,
                0, MPI_COMM_WORLD);

    // Local sort
    quicksort(local_data, 0, local_n - 1);
```

Ln 65, Col 52 Spaces: 4

Welcome | mpi_quicksort.cpp

mpi_quicksort.cpp

```cpp
int main(int argc, char* argv[]) {

    // Local sort
    quicksort(local_data, 0, local_n - 1);

    // Gather sorted subarrays
    vector<int> gathered;
    if (rank == 0) {
        gathered.resize(n);
    }

    MPI_Gather(local_data.data(), local_n, MPI_INT,
               gathered.data(), local_n, MPI_INT,
               0, MPI_COMM_WORLD);

    // Final merge at root
    if (rank == 0) {
        sort(gathered.begin(), gathered.end());

        cout << "Sorted array: ";
        for (int x : gathered)
            cout << x << " ";
        cout << endl;
    }

    MPI_Finalize();
    return 0;
}
```

Ln 81, Col 2 Spaces: 4 U

```
    Topology libraries available:          hwloc
    Resource management kernels available:   user slurm ll lsf sge pbs cobalt
    Checkpointing libraries available:
    Demux engines available:               poll select
rutu@Rutuja:/mnt/c/Users/dell/Videos/BE Practicals/HPC$ mpicxx mpi_quicksort.cpp -o mpi_quicksort
rutu@Rutuja:/mnt/c/Users/dell/Videos/BE Practicals/HPC$ mpirun -np 4 ./mpi_quicksort
Original array: 83 86 77 15 93 35 86 92 49 21 62 27 90 59 63 26
Sorted array: 15 21 26 27 35 49 59 62 63 77 83 86 86 90 92 93
rutu@Rutuja:/mnt/c/Users/dell/Videos/BE Practicals/HPC$
```

23°C Sunny Search

CONCLUSION

Quicksort is an efficient sorting algorithm with excellent average-case performance. However, its sequential nature limits its efficiency on large datasets. By incorporating MPI-based parallelization, the sorting process can be distributed across multiple processes, significantly reducing execution time.

The project demonstrates that parallel Quicksort is more effective for large datasets, while sequential Quicksort remains suitable for smaller inputs. This study highlights the importance of parallel computing in improving algorithm performance and utilizing system resources efficiently.

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